Emiliano Paradiso

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Summary

Curious, hands-on and strongly motivated in the field of robotics, I'm a Master's student close to graduation and currently looking for my first opportunity in the sector. I enjoy working in collaborative environments, sharing ideas and learning from feedback. I'm the kind of person who prefers learning by doing, experimenting, breaking things and fixing them better than before.
If you'd like to get a better idea of how I work, you can have a look at my projects on my GitHub page!
https://github.com/Emilianogith

Skills

- ROS/ROS2
- C++
- MATLAB / Simulink Robotics
- Python
- Deep learning
- Reinforcement learning
- Computer vision
- Gazebo / MuJoCo / MuJoCo Playground / CoppeliaSim
- Shared control strategies for collaborative robots
- Optimal control and trajectory optimization
- State estimation
- Motion planning
- Linux development environment (Ubuntu)
- Git
- Teamwork and technical communication in multidisciplinary projects

Education

Master's degree : **Artificial Intelligence and Robotics**
2023 to Current
University of Rome "La Sapienza" - Rome

- During my academic path I have worked on several projects in motion planning, control, perception, localization and mapping for mobile and autonomous robots, both in simulation and on real platforms, individually and in teams. Through the Artificial Intelligence specialization track I have developed a strong interdisciplinary skill set, integrating AI methods into my robotics expertise.
- I am currently completing my Master's thesis in the university research lab, where I bring together the multidisciplinary knowledge acquired during my studies to develop an autonomous high-speed obstacle-jumping controller for a wheeled mobile robot.

Bechelor's degree : **Electronic engineering**
2020 to 2023
University of Rome "La Sapienza" - Rome

- During my Bachelor's degree I acquired a solid theoretical foundation in electronics, including analog and digital electronics, electromagnetic fields, signal theory, antennas, sensors, actuators and information technology. This knowledge was reinforced through several hands-on laboratory activities and practical projects.
- 110/110 cum laude Graduate

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Languages

Italian: First Language
English: B2
Upper Intermediate

Certifications

- I-RIM - 5a Conferenza Italiana di Robotica e Macchine Intelligenti *Roma, 20-22 ottobre 2023.*
- Fusion/Fusion 360 3D Modeling.
- Regione Lazio – Faberlab course on designing and building antweight combat robots for Maker Faire

Projects

- Dynamic Obstacle jumping on TITA wheeled mobile robot
- MPC for dynamic locomotion in the MIT cheetah3 robot
- Reinforcement learning Hindsight Goal Ranking for pick and place operation tasks
- Manifold Least Square mobile robot calibration
- MOD-RRT for real time planning in dynamic environments
- Noisy graph classification with Graph Isomorphism Network
- Shared human–robot control for teleoperated ultrasound imaging
- Robust control of an industrial robot arm
- Stubborn consensus protocol or multiagent systems
- Learning Variable Whole-Body Control for Agile Aerial Manipulation in Strong Winds